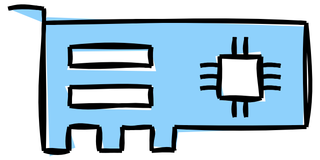
Choosing a GPU

First....
Do you even need a GPU?



Dedicated GPU

I want to own and play the latest AAA games

The latest and greatest releases on Console / PC from big budget studios

I run several or large screen monitors 1440p, 4k, Ultrawide, 3x 1080p

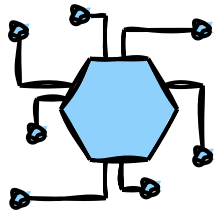
I work on creating and editing video / photos

Photoshop benefits from a GPU

My main use app can benefit from a GPU

e.g. Crypto mining, blockchain, machine learning (AI)...

The choice of most serious users....



Integrated GPU or iGPU

I casually game and play undemanding games

Windows store games

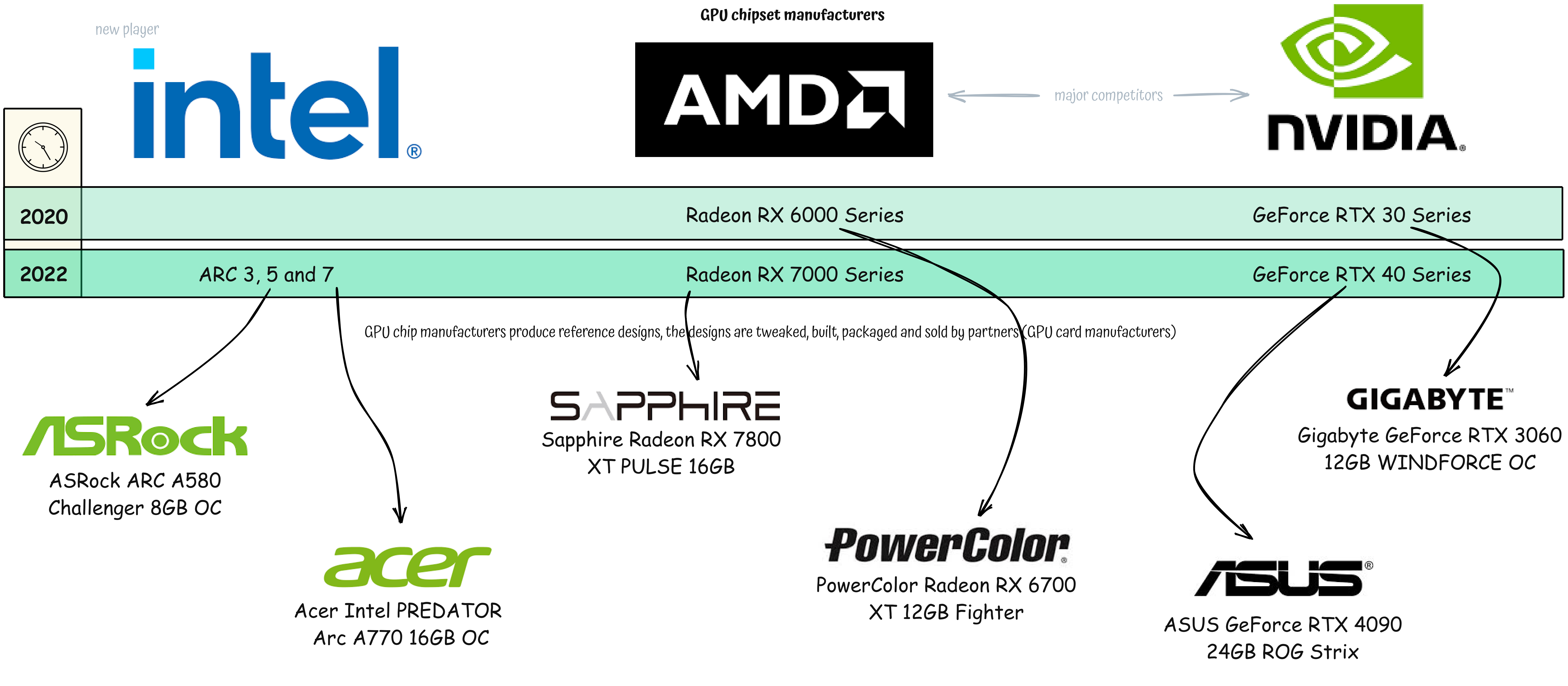
I'm happy to run games on low settings

Fortnite and GTA V can run on LOW settings

I'm on a tight budget

These are getting better all the time, but we're a long way away from them matching a dedicated GPU for pure performance

The GPU Market



Common Manufacturer Tweaks > More Memory (GB), Overclocked (OC), Better Cooling (2 or 3 fan variants), LED's

GPU Chips / Models No's

Gamers, Professionals and Creators

Series	Models	Introduced
Nvidia GeForce RTX 40	4090 4080 4070 Ti 4070 4060 Ti 4060	Oct 2022 – Jun 2023
AMD Radeon RX 7000	7900 XTX 7900 XT 7900 GRE 7800 XT 7700 XT 7600	Sep 2022 – Sep 2023
Intel ARC Gen 12.7	A770 16GB A770 8GB A750 A580 A380 A310	Jun 2022 – Oct 2023

Nvidia Ti = Titanium

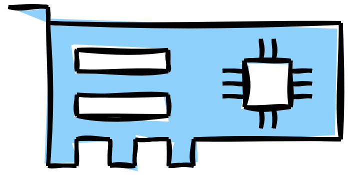
Series	Models	Introduced
Nvidia GeForce RTX 30	3090 Ti 3090 3080 Ti 3080 3070 Ti 3070 3060 Ti 3060 3050	Dec 2020 – Dec 2022
AMD Radeon RX 6000	6950 XT 6900 XT 6800 XT 6800 6750 XT 6700 XT 6700 6650 XT 6600 XT 6600 6500 XT 6400	Aug 2021 – May 2022

AMD XT = Extreme

AMD XTX = More Extreme

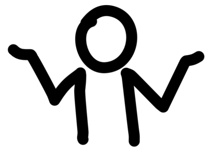
GPU Model No's

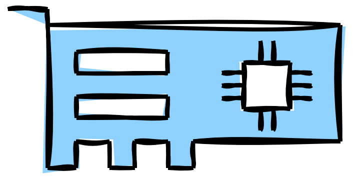
Science and Serious Media (Pro's)



Range	Models	Introduced
RTX Ada	6000 ($4250) 5000 4500 4000 ($999)	Dec 2022 – Mar 2023
Radeon PRO 7000	W7900 ($3999) W7800 ($2499) W7600 ($599) W7500 ($429)	Jan 2023 – Aug 2023
RTX A	A6000 ($4000) A5500 A5000 A4500 ($960) A4000 A2000	Oct 2020 – Mar 2022
Radeon PRO 6000	W6800 ($2249) W6600 ($649) W6400 ($229)	Jun 2021 – Oct 2022

Which GPU Chipset Should I Choose?





What do you want to power?

Name	Resolution	Pixels
1080p or Full HD	1920×1080	2 million
1440p	2540×1440	3.6 million
WQHD UltraWide	3440×1440	4.9 million
4k	2840×2160	6.1 million
DQHD UltraWide	5120×1440	7.3 million
8k	7680×4320	33.1 million

What (FPS) frames per second?

PS5 = 60 to 120 fps, Movie = 24 fps

General gaming works at 30-60 fps, competitive? 120+

What's the monitor refresh rate (Hz)?

60 was common, now 75Hz, 120Hz, 144Hz, 240Hz (and higher) are available

e.g. Modern TV.... LG - QNED (2023) has 120Hz refresh rate

Monitor refresh rate should be the same if not higher than FPS

Which GPU? >

Minimum GPU Chipset	FPS / Quality	Approx Cost (Oct 2023)
Nvidia RTX 4090 AMD RX 7900 XTX	100 fps ultra quality 4k 140 fps ultra quality 1440p 160 fps ultra quality 1080p 190 fps medium quality 1080p	$900-$1800
Nvidia RTX 4080 AMD RX 7900 XT	80 fps ultra quality 4k 120 fps ultra quality 1440p 140 fps ultra quality 1080p 180 fps medium quality 1080p	$800-$1300
Nvidia RTX 4070 AMD RX 7800 XT Nvidia RTX 3080 AMD RX 6800 XT	60 fps ultra quality 4k 100 fps ultra quality 1440p 120 fps ultra quality 1080p 170 fps medium quality 1080p	$450-$550
Nvidia RTX 4060 Ti AMD RX 7700 XT Nvidia RTX 3070 Ti	40 fps ultra quality 4k 75 fps ultra quality 1440p 100 fps ultra quality 1080p 160 fps medium quality 1080p	$400
Nvidia RTX 4060 AMD RX 7600 Intel ARC A770 16GB Nvidia RTX 3060 AMD RX 6600	30 fps ultra quality 4k 60 fps ultra quality 1440p 80 fps ultra quality 1080p 140 fps medium quality 1080p	$200-$300
Intel ARC A380 AMD RX6400	< 20 fps ultra quality 1440p 25 fps ultra quality 1080p 50 fps medium quality 1080p	$125

AI Upscaling and Power Requirements

DLSS and FSR

Deep Learning Super Sampling

FideityFX Super Resolution

Better frame rates on supported games

AI to upscale lower resolution images

DLSS only on Nvidia RTX 40 Series and newer

Possibly 3000 series with tweaks

FSR on Nvidia, Intel or AMD

Which is better?

That's a game and user specific question!!

POWER

Modern high end cards are power hungry

Often require multiple 6 or 8 pin connections (check PSU requirements)

ATX 3.0 new spec PSU's have a 12+4 pin connector for GPU's

	AMD Radeon™ RX 7900 XTX	
GPU	Compute Units: 96	Boost Frequency: Up to 2500 MHz
	Ray Accelerators: 96	AI Accelerators: 192
	Peak Texture Fill-Rate: Up to 960 GT/s	Peak Half Precision Compute Performance: 123 TFLOPs
	ROPs: 192	Stream Processors: 6144
	Transistor Count: 58 B	
Requirements	Typical Board Power (Desktop): 355 W	Minimum PSU Recommendation: 800 W

Size and Ports

Will it fit?

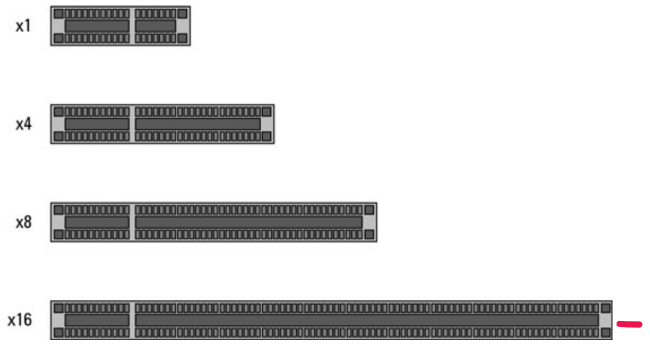
Most take a PCI-E x16 slot

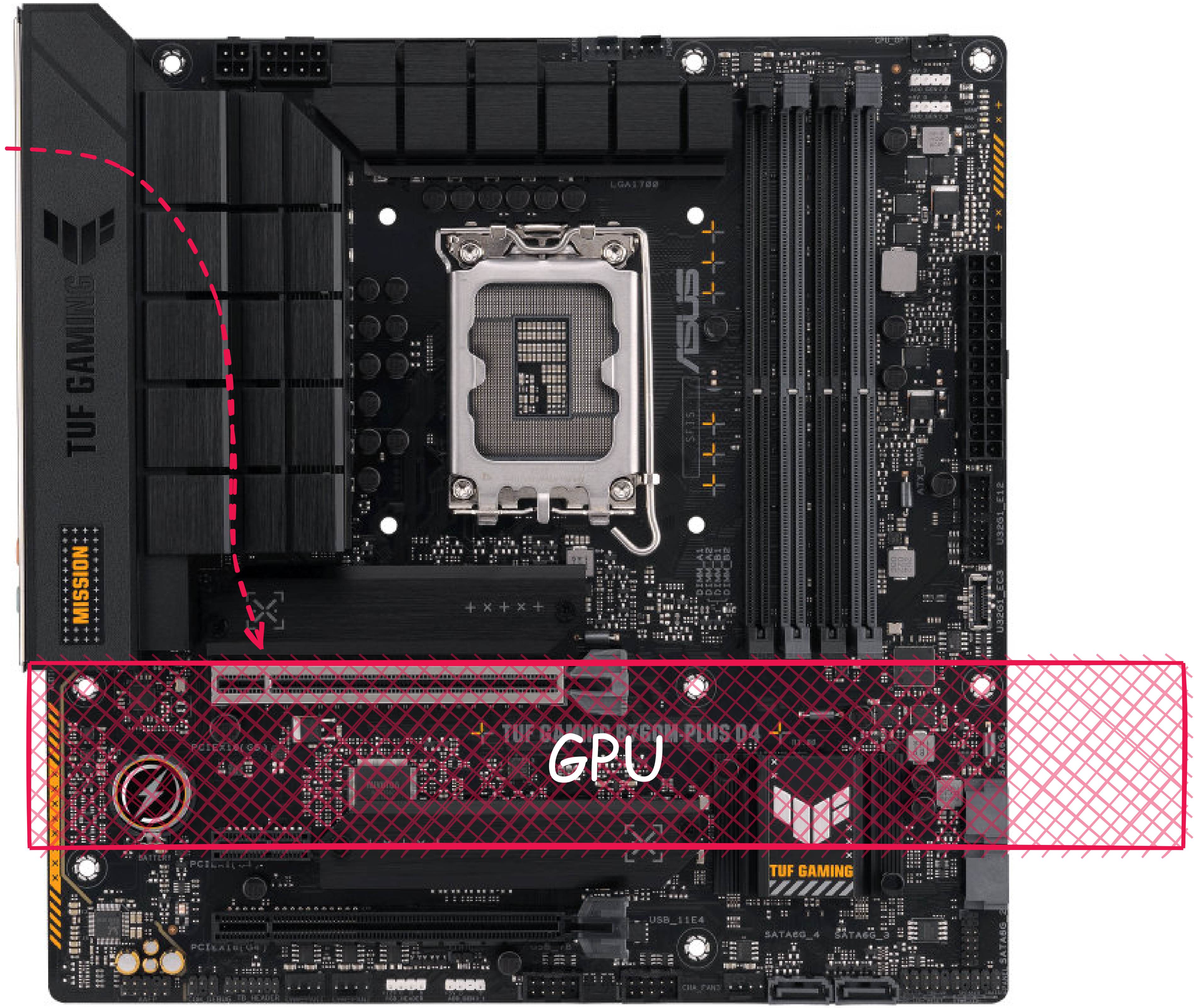
Some high end cards are very large

Cover 2 or 3 PCI-E slots (slot width)

Long and can sag (backplates are added, supports are available)

Check case clearance





Micro-ATX Mobo

Multi Monitors - Check Ports...

HDMI - how many? (x1)

Display Port - how many? (x3)

Can an Intel CPU work with an AMD GPU?
Or an AMD CPU with an Intel GPU?

Good question...

but don't worry...

they're all compatible...

Recommendations

Figure out your available budget

Figure out what you want to power (size and Hz) and your acceptable FPS

!! Monitor refresh rate should be the same if not higher than FPS !!

Casual competitive gamer
60-120 fps on 1440p 120Hz Monitor

Find the GPU chipsets that power that resolution well

Look for reviews of various GPU chipsets in your game (choose a chipset)

"RTX 4070 vs RX 7800 XT Fortnite"

Shop for manufacturer cards with that chipset and look for reviews of those cards

"PowerColor Radeon RX 7800 XT Review"

Don't be swayed by OC models (you can do this yourself)

Check card size, power, ports then.... commit!

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